HK IMO Training 2021-2022
Problem Set 15 Suggested Solutions

Problem 57 (Thailand MO 2015 Problem 1)**.** Let p be a prime number. Let $a_1, a_2, \ldots$ be a sequence of positive integers such that

$$a_{n+2} = \frac{a_{n+1}^2 + p}{a_n}$$

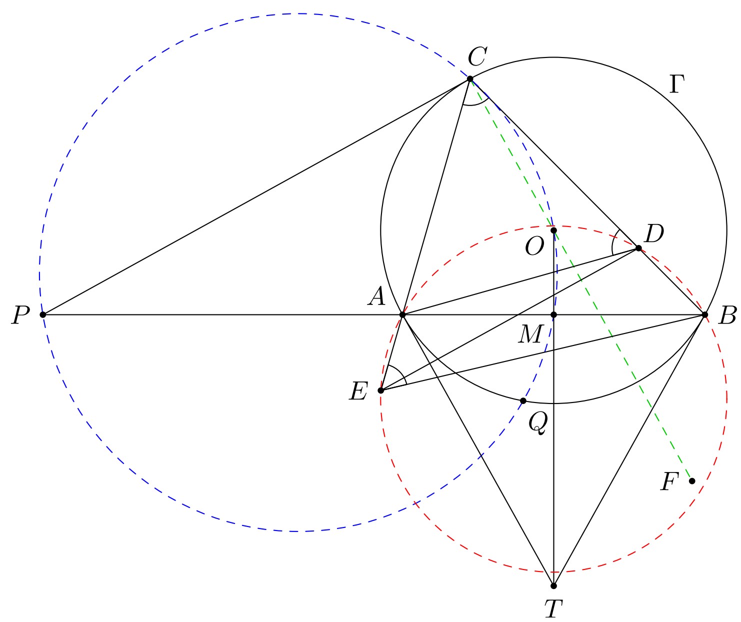
for $n = 1, 2, \ldots$. Prove that a_{n+1} divides $a_n + a_{n+2}$ for any positive integer n.

Solution. Rewrite the equation as $a_n a_{n+2} - a_{n+1}^2 = p$. Replacing n by $n+1$ and comparing with this equation, we obtain $a_n a_{n+2} - a_{n+1}^2 = a_{n+1}a_{n+3} - a_{n+2}^2$. This implies

$$a_{n+2}(a_n + a_{n+2}) = a_{n+1}(a_{n+1} + a_{n+3}).$$

Thus, a_{n+1} divides $a_{n+2}(a_n + a_{n+2})$. We are done if we can prove that $(a_{n+1}, a_{n+2}) = 1$. Suppose $(a_{n+1}, a_{n+2}) = d$. Then d divides $a_n a_{n+2} - a_{n+1}^2 = p$. If $d = p$, by $a_{n+3}^2 = a_{n+2}a_{n+4} - p$, we know that p divides a_{n+3}. But then p^2 divides $a_{n+1}a_{n+3} - a_{n+2}^2 = p$, contradiction. So $d = 1$, and the proof is complete. $\square$

Problem 58 (Vietnam MO 2014 Problem 5)**.** Let A and B be two points on a circle Γ. Consider a variable point C on the major arc of AB on Γ. Let $D \neq C$ be the point on the line BC such that $AC = AD$. Let $E \neq C$ be the point on the line AC such that $BC = BE$. The circle with centre D and passing through C meets the circle with centre E and passing through C again at F. The line passing through C and perpendicular to CF intersects the line AB at P. Let M be the midpoint of AB. The circumcircle of $\triangle CPM$ meets Γ again at Q. Prove that CQ passes through a fixed point independent of C.



Solution. Let T be the intersection of the tangents at A and B to Γ, which is independent of C. We claim that T always lies on CQ. Firstly, since $\measuredangle CDA = \measuredangle ACD = \measuredangle ECB = \measuredangle BEC$, the points A, E, B, D are concyclic. Note that CF is perpendicular to the line DE joining the centres of the two circles. Thus, CP is parallel to DE. This implies $\measuredangle PCA = \measuredangle DEC = \measuredangle CBA$, and hence PC is tangent to Γ.

Let O be the centre of Γ. Then $\angle PCO = 90°$ and $\angle PMO = 90°$. This shows O lies on $(CPQM)$. Note that T lies on the line OM. As $\triangle TAM \sim \triangle TOA$, we have $TA^2 = TM \times TO$. This implies T has the same power with respect to $(CPQMO)$ and Γ. Thus, T lies on the radical axis CQ of these circles as desired. $\square$

Problem 59 (ARMO 2011 Grade 11 Problem 3). Let $S_1, S_2, \dots, S_{999}$ be subsets of S such that $|S_i \cap S_j| = 1$ for any $i \neq j$. Suppose for every $x \in S$, there are exactly three subsets S_i containing x. Prove that there is a subset T of S such that $|T| = 250$ and $|T \cap S_i| \leqslant 1$ for any i.

Solution. Consider any subset T of S satisfying $|T \cap S_i| \leqslant 1$ for any i, for which $|T| = n$ is maximized. We are done if $n \geqslant 250$. Suppose on the contrary that $n \leqslant 249$. Let $T = \{x_1, x_2, \dots, x_n\}$. Since no S_j contains more than one element in T, we may assume $x_i \in S_{3i-2}, S_{3i-1}, S_{3i}$ for $i = 1, 2, \dots, n$.

Due to the maximality of n, no three sets among $S_{3n+1}, S_{3n+2}, \dots, S_{999}$ share the same element. Since each pair of these sets share exactly one common element, there are $\binom{999-3n}{2}$ elements belonging to two of $S_{3n+1}, S_{3n+2}, \dots, S_{999}$ and one of $S_1, S_2, \dots, S_{3n}$. We call these special elements.

Claim: For each $i = 1, 2, \dots, n$, $S_{3i-2} \cup S_{3i-1} \cup S_{3i}$ contains at most $\frac{999-3n}{2}$ special elements.

Proof. Suppose on the contrary that $S_1 \cup S_2 \cup S_3$, for example, contains more than $\frac{999-3n}{2}$ special elements. Note that these special elements each belongs to exactly one of S_1, S_2, S_3 by definition. If a special element belongs to S_1, S_{i_1}, S_{i_2} and another special element belongs to S_1, S_{j_1}, S_{j_2}, then i_1, i_2, j_1, j_2 are pairwise distinct (since any two sets share only one common element). Thus, each special element belonging to S_1 must belong to two different sets among $S_{3n+1}, S_{3n+2}, \dots, S_{999}$. So there are at most $\frac{999-3n}{2}$ special elements belonging to S_1.

Therefore, we may assume both S_1 and S_2 contain some special elements. If a special element y belongs to S_1, S_{i_1}, S_{i_2} and another special element z belongs to S_2, S_{j_1}, S_{j_2}, then one of i_1, i_2 must be equal to one of j_1, j_2, since otherwise we can remove x_1 from T and add back y and z to T so that the conditions are satisfied but $|T|$ is increased by 1, contradicting the maximality of n. As a corollary, S_1 contains at most 2 special elements (if there are 3 special elements, one belonging to S_1, S_{i_1}, S_{i_2}, one belonging to S_1, S_{i_3}, S_{i_4}, and one belonging to S_1, S_{i_5}, S_{i_6}, then one of $\{i_1, i_2\}$, $\{i_3, i_4\}$ and $\{i_5, i_6\}$ does not contain j_1 and j_2). By symmetry, S_2 and S_3 each contains at most 2 special elements. Thus, $S_1 \cup S_2 \cup S_3$ contains at most $6 < \frac{999-3n}{2}$ special elements, contradiction. □

By the claim, we see that there are at most $\frac{n(999-3n)}{2}$ special elements. This gives $\binom{999-3n}{2} \leqslant \frac{n(999-3n)}{2}$, which means $(999-3n)(998-3n) \leqslant n(999-3n)$. This implies $998 \leqslant 4n$, contradicting $n \leqslant 249$. So we are done. □

Problem 60 (Canada MO 2019 Problem 4). Let $n \geqslant 3$ be a positive integer, and let $a_1, a_2, \dots, a_n, c$ be real numbers such that $a_2 = a_{n-1} = 0$. Prove that

$$\sum_{j=1}^{n-2} |a_j - ca_{j+1} - a_{j+2}| \geqslant |a_1| - |a_n| .$$

Solution. When $n = 3$, we need to prove $|a_1 - a_3| \geqslant |a_1| - |a_3|$, which holds by the triangle inequality. In the following, assume $n \geqslant 4$.

The idea is to choose some weights $w_1, w_2, \dots, w_{n-2}$ such that $w_1 = 1$ and $|w_i| \leqslant 1$ for $2 \leqslant i \leqslant n-2$, and $w_j = cw_{j-1} + w_{j-2}$ for $3 \leqslant j \leqslant n-2$. Then by the triangle inequality, we have

$$\begin{aligned}
\sum_{j=1}^{n-2} |a_j - ca_{j+1} - a_{j+2}| &\geqslant \sum_{j=1}^{n-2} |w_j|\,|a_j - ca_{j+1} - a_{j+2}| \\
&\geqslant \left| \sum_{j=1}^{n-2} (w_j(a_j - ca_{j+1} - a_{j+2})) \right| \\
&= \left| w_1 a_1 - w_{n-2} a_n + \sum_{j=3}^{n-2} (-w_{j-2} - cw_{j-1} + w_j) a_j \right| \\
&= |a_1 - w_{n-2} a_n| \\
&\geqslant |a_1| - |w_{n-2} a_n| \\
&\geqslant |a_1| - |a_n| .
\end{aligned}$$

It remains to show that such weights exist.

In view of the recurrence relation $w_j = cw_{j-1} + w_{j-2}$, we may consider $w_j = \alpha^{j-1}$ where $\alpha = \frac{c \pm \sqrt{c^2+4}}{2}$ is the root of the characteristic equation $\lambda^2 - c\lambda - 1 = 0$ for which $|\alpha| \leqslant 1$ (which exists since the product of roots is -1). Then the recurrence relation holds, $w_1 = 1$ and $|w_i| \leqslant 1$ for $2 \leqslant i \leqslant n-2$. So we are done. □